Assignment - 8

Task - 1:

```
from ultralytics import YOLO

model= YOLO("yolov8n.pt")

results= model.train(
      data="dat/data.yaml",
      epochs=50,
      imsz=640,
      batch=8,
      augment=True,
)

# Evalute:

metrics = model.val()

results= model.predict(source="data/images/test, conf=0.25)

print("mAP50:", metrics.map50)

print("mAP50-95:", metrics.box.map)
```

Report:

This task involved fine-tuning a YOLOv8 model to detect computer monitors - a real world object that pre-trained models often fail to recognize accurately. A custom dataset of 35 images captured using a mobile phone under varying ~~ligth~~ lighting, angles. The dataset was ~~test~~ split into 70:20:10 ration and images were resized 640x640px

Model:

Base Model: YOLOv8n pre-trained on coco dataset

Experimental Results:

Validation Performance:

Precision: 0.891

Recall: 0.923

mAP@50: 0.912

mAP@50-95: 0.852

Inference Speed (CPU) ≃222ms per image

Conclusion: The fine-tuned YOLOv8n model demonstrated strong performance in detecting computer monitors in real world scenarios.

Task 2.1: Object detection in video

```
from ultralytics import YOLO
import time

video_path = "/home/saidul/data/video.mp4"

models = {
        "YOLOv8n": "yolov8n.pt",
        "YOLOv11n": "yolo11n.pt",
        "YOLOv12n": "yolo12n.pt"
}
results_summary = {}
for model_name, weight in models.items():
        mode = YOLO(weight)
        start_time = time.time()
        results = model.predict(
            source = video_path,
            save = True,
            conf = 0.25,
            stream = True
            verbose = False
        }
```

```
end_time = time.time()
total_time = end_time - start_time

frame_count = 0
for _in results:
    frame_count += 1
fps = frame_count / total_time if total_time > 0 else 0

results_summary[model_name] = {"time": total_time,
                               "fps": fps}
print("summary")
for name, data in results_summary.items():
    print(f"{name} → Time: {data['time']}s ~~FPS: {data['fps~~
         | FPS: {data['fps']:.1f}")
```

Report: A 15 second video was captured using a smartphone in an indoor environment.

Experimental Results

Model	Total Time to finish	Apprx. FPS	Observation
YOLOv8n	0.03 seconds	14,762	Fastest inference
YOLOv11n	0.07 sec	6,463	Slowest among three
YOLOv12n	0.05 sec	9,599	Good bal…

Analysis: YOLO v8n provided the best inference speed while YOLOv11n and YOLO v12n offered better detection quality at the cost of speed.

Task 2.2 : Widen Face Face detecton

```
import time
import torch
import cv2
import matplotlib.pyplot as pt
import numpy as np
import torch.nn as nn
from ultralytics import YOLO

dataset_root = "/kaggle/input/widen-face-for-yolo-training"

data-yaml = {
        'path': dataset-root,
        'train': 'images/train',
        'val' : 'images/val'
        'names': {0:'face'}

  }
with open ('/kaggle/working/data.yaml', 'w')as f:
        yaml.dump(data-yaml, f)
```

```
model-v8 = YOLO ('yolov8n.pt')

results-v8 = model-v8.train(
        data='/kaggle/working/data.yaml',
        epochs=25,
        imgsz=640,
        batch=16,
        device=0,
        project='/kaggle/working/yolov8-face-detection',
)

val-metrics = model-v8.val

github-model = YOLO('/kaggle/working/yolov8n-face.pt')
github-metrics = github-model.val(data='/kaggle/working/data.yaml'
print (f{github-metrics.box.mp: .4f}")

print (f{github-metrics.box.map50: .4f}")
```

Report:

Dataset: WIDER FACE dataset (subset used due to hardware limitations).

Mode: YOLOv8n fine tuned on the data set

Training details:

Epochs: 25

Batch Size: 4

image-size: 640x640

A simple YOLOv1-style face detector was designed and implemented. The model was trained on the same WIDER FACE training subset.

~~Perform~~ The YOLOv1-based model was much simpler but struggled with small faces. Modern YOLO v8 models significantly outperform the classic YOLOv1 approach due to better architecture, anchor boxes and advanced training techniques

Task-1

Fine-tune an object detection model such as YOLO to detect and classify custom real-world objects captured using mobile devices that are not adequately recognized by pre-trained models.

To complete the assignment, perform the following tasks:

Collect a custom image dataset containing the target objects captured under different real-world conditions such as varying illumination, background complexity, viewing angles, occlusion, and object scales.

Annotate the dataset using appropriate bounding-box annotations and divide it into training, validation, and testing sets.

Apply suitable data augmentation techniques to improve the robustness and generalization capability of the model.

Fine-tune a pre-trained object detection model on the custom dataset.

Evaluate the performance of the fine-tuned model using standard object detection metrics, including:

Precision

Recall

Mean Average Precision (mAP)

Inference speed and detection latency

Analyze the strengths and limitations of the developed detector and compare its performance with the original pre-trained model on the custom object categories.

The report should include:

Dataset preparation and annotation methodology

Model architecture and training configuration

Data augmentation strategies

Experimental results and performance analysis

Qualitative detection examples

Discussion and conclusion

In [ ]:

```
from ultralytics import YOLO
import os

# Load pre-trained YOLOv8n model
model = YOLO("yolov8n.pt")   # Nano version - best for CPU

# Fine-tune
results = model.train(
    data="/home/saidul/Desktop/4-2/Deep Learning/data/monitor_detection/data
    epochs=50,                   # You can increase to 100 later
    imgsz=640, # Standard size for YOLOv8: 640x640 px
    batch=8,                # Small batch for CPU
    name="monitor_detector",
    patience=20,
    augment=True,
    device='cpu'
)

print("Training completed!")
```

```
Ultralytics 8.4.51 🚀 Python-3.12.3 torch-2.12.0+cu130 CPU (Intel Core i7-86
65U 1.90GHz)
engine/trainer: agnostic_nms=False, amp=True, angle=1.0, augment=True, auto_
augment=randaugment, batch=8, bgr=0.0, box=7.5, cache=False, cfg=None, class
es=None, close_mosaic=10, cls=0.5, cls_pw=0.0, compile=False, conf=None, cop
y_paste=0.0, copy_paste_mode=flip, cos_lr=False, cutmix=0.0, data=/home/said
ul/Desktop/4-2/Deep Learning/data/monitor_detection/data.yaml, degrees=0.0,
deterministic=True, device=cpu, dfl=1.5, dnn=False, dropout=0.0, dynamic=Fal
se, embed=None, end2end=None, epochs=50, erasing=0.4, exist_ok=False, fliplr
=0.5, flipud=0.0, format=torchscript, fraction=1.0, freeze=None, half=False,
hsv_h=0.015, hsv_s=0.7, hsv_v=0.4, imgsz=640, int8=False, iou=0.7, keras=Fal
se, kobj=1.0, line_width=None, lr0=0.01, lrf=0.01, mask_ratio=4, max_det=30
0, mixup=0.0, mode=train, model=yolov8n.pt, momentum=0.937, mosaic=1.0, mult
i_scale=0.0, name=monitor_detector, nbs=64, nms=False, opset=None, optimize=
False, optimizer=auto, overlap_mask=True, patience=20, perspective=0.0, plot
s=True, pose=12.0, pretrained=True, profile=False, project=None, rect=False,
resume=False, retina_masks=False, rle=1.0, save=True, save_conf=False, save_
crop=False, save_dir=/home/saidul/Desktop/4-2/Deep Learning/runs/detect/moni
tor_detector, save_frames=False, save_json=False, save_period=-1, save_txt=F
alse, scale=0.5, seed=0, shear=0.0, show=False, show_boxes=True, show_conf=T
rue, show_labels=True, simplify=True, single_cls=False, source=None, split=v
al, stream_buffer=False, task=detect, time=None, tracker=botsort.yaml, trans
late=0.1, val=True, verbose=True, vid_stride=1, visualize=False, warmup_bias
_lr=0.1, warmup_epochs=3.0, warmup_momentum=0.8, weight_decay=0.0005, worker
s=8, workspace=None
Downloading https://ultralytics.com/assets/Arial.ttf to '/home/saidul/.confi
g/Ultralytics/Arial.ttf': 100% ━━━━━━━━━━━━ 755.1KB 2.5MB/s 0.3s 0.2s<0.2s3s
Overriding model.yaml nc=80 with nc=1

                   from  n    params  module
arguments
  0                  -1  1       464  ultralytics.nn.modules.conv.Conv
[3, 16, 3, 2]
  1                  -1  1      4672  ultralytics.nn.modules.conv.Conv
[16, 32, 3, 2]
  2                  -1  1      7360  ultralytics.nn.modules.block.C2f
[32, 32, 1, True]
  3                  -1  1     18560  ultralytics.nn.modules.conv.Conv
[32, 64, 3, 2]
  4                  -1  2     49664  ultralytics.nn.modules.block.C2f
[64, 64, 2, True]
  5                  -1  1     73984  ultralytics.nn.modules.conv.Conv
[64, 128, 3, 2]
  6                  -1  2    197632  ultralytics.nn.modules.block.C2f
[128, 128, 2, True]
  7                  -1  1    295424  ultralytics.nn.modules.conv.Conv
[128, 256, 3, 2]
  8                  -1  1    460288  ultralytics.nn.modules.block.C2f
[256, 256, 1, True]
  9                  -1  1    164608  ultralytics.nn.modules.block.SPPF
[256, 256, 5]
 10                  -1  1         0  torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']
 11             [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 12                  -1  1    148224  ultralytics.nn.modules.block.C2f
```

```
[384, 128, 1]
 13                  -1  1         0  torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']
 14             [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 15                  -1  1     37248  ultralytics.nn.modules.block.C2f
[192, 64, 1]
 16                  -1  1     36992  ultralytics.nn.modules.conv.Conv
[64, 64, 3, 2]
 17            [-1, 12]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 18                  -1  1    123648  ultralytics.nn.modules.block.C2f
[192, 128, 1]
 19                  -1  1    147712  ultralytics.nn.modules.conv.Conv
[128, 128, 3, 2]
 20             [-1, 9]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 21                  -1  1    493056  ultralytics.nn.modules.block.C2f
[384, 256, 1]
 22        [15, 18, 21]  1    751507  ultralytics.nn.modules.head.Detect
[1, 16, None, [64, 128, 256]]
Model summary: 130 layers, 3,011,043 parameters, 3,011,027 gradients, 8.2 GF
LOPs

Transferred 319/355 items from pretrained weights
Freezing layer 'model.22.dfl.conv.weight'
train: Fast image access ✅ (ping: 0.0±0.0 ms, read: 128.4±44.1 MB/s, size:
107.1 KB)
train: Scanning /home/saidul/Desktop/4-2/Deep Learning/data/monitor_detectio
n/labels/train... 23 images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 23/
23 745.1it/s 0.0s
train: New cache created: /home/saidul/Desktop/4-2/Deep Learning/data/monito
r_detection/labels/train.cache
val: Fast image access ✅ (ping: 0.0±0.0 ms, read: 142.8±40.8 MB/s, size: 10
1.1 KB)
val: Scanning /home/saidul/Desktop/4-2/Deep Learning/data/monitor_detection/
labels/val... 6 images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 6/6 880.
3it/s 0.0s
val: New cache created: /home/saidul/Desktop/4-2/Deep Learning/data/monitor_
detection/labels/val.cache
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937'
and determining best 'optimizer', 'lr0' and 'momentum' automatically...
optimizer: AdamW(lr=0.002, momentum=0.9) with parameter groups 57 weight(dec
ay=0.0), 64 weight(decay=0.0005), 63 bias(decay=0.0)
Plotting labels to /home/saidul/Desktop/4-2/Deep Learning/runs/detect/monito
r_detector/labels.jpg...
Image sizes 640 train, 640 val
Using 0 dataloader workers
Logging results to /home/saidul/Desktop/4-2/Deep Learning/runs/detect/monito
r_detector
Starting training for 50 epochs...

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
       1/50         0G     0.9581      2.971       1.13         49        64
0: 100% ━━━━━━━━━━━━ 3/3 6.2s/it 18.6s<10.0s
```

```
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.4s/it 1.4s
                   all          6         13    0.00667      0.923     0.091
7     0.0538

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
       2/50         0G     0.7098      2.656     0.9927         31        64
0: 100% ━━━━━━━━━━━━ 3/3 4.3s/it 12.9s7.8ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.2s/it 1.2s
                   all          6         13    0.00722          1      0.36
6      0.309

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
       3/50         0G      0.634      2.358     0.9749         18        64
0: 100% ━━━━━━━━━━━━ 3/3 4.0s/it 11.9s7.0ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.4s/it 1.4s
                   all          6         13    0.00722          1      0.57
7      0.525

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
       4/50         0G     0.5829      1.777     0.9234         29        64
0: 100% ━━━━━━━━━━━━ 3/3 4.9s/it 14.7s8.7s7s
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.7s/it 1.7s
                   all          6         13    0.00722          1      0.64
5      0.595

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
       5/50         0G       0.48      1.533     0.9184         32        64
0: 100% ━━━━━━━━━━━━ 3/3 4.7s/it 14.1s8.3s5s
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.1s/it 1.1s
                   all          6         13      0.283      0.615      0.64
9      0.593

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
       6/50         0G     0.5264      1.238     0.9169         35        64
0: 100% ━━━━━━━━━━━━ 3/3 4.6s/it 13.9s8.9ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.2s/it 1.2s
                   all          6         13          1      0.202      0.66
3      0.619

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
       7/50         0G     0.5215      1.182      0.897         27        64
0: 100% ━━━━━━━━━━━━ 3/3 3.7s/it 11.1s6.5ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.1s/it 1.1s
```

```
                   all          6         13      0.622      0.133      0.69
3      0.583

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
       8/50         0G     0.5692      1.088     0.9037         39         64
0: 100% ━━━━━━━━━━━━ 3/3 3.9s/it 11.8s7.4ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.6s/it 1.6s
                   all          6         13      0.799      0.307      0.62
2      0.534

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
       9/50         0G     0.5133      1.026     0.9025         41         64
0: 100% ━━━━━━━━━━━━ 3/3 4.5s/it 13.6s8.0ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.3s/it 1.3s
                   all          6         13      0.808      0.385      0.59
6      0.523

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      10/50         0G     0.5662      1.024     0.8727         49         64
0: 100% ━━━━━━━━━━━━ 3/3 3.9s/it 11.7s7.5ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.8s/it 1.8s
                   all          6         13      0.842      0.615      0.63
5      0.585

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      11/50         0G     0.5405     0.9506     0.9083         34         64
0: 100% ━━━━━━━━━━━━ 3/3 3.7s/it 11.0s7.0ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.1s/it 1.1s
                   all          6         13      0.879       0.56      0.63
8      0.589

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      12/50         0G     0.4899      1.054     0.8961         35         64
0: 100% ━━━━━━━━━━━━ 3/3 3.5s/it 10.5s6.6ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.1s/it 1.1s
                   all          6         13      0.784      0.561      0.69
1      0.596

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      13/50         0G     0.5935      1.045     0.9298         22         64
0: 100% ━━━━━━━━━━━━ 3/3 3.8s/it 11.4s6.9ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.4s/it 1.4s
                   all          6         13      0.935      0.692      0.74
1       0.61
```

```
      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      14/50         0G     0.5255     0.9239      0.879         32        64
0: 100% ━━━━━━━━━━━━ 3/3 3.5s/it 10.5s6.5ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.2s/it 1.2s
                   all          6         13      0.949      0.692      0.74
7      0.643

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      15/50         0G     0.5508     0.9475     0.9047         43        64
0: 100% ━━━━━━━━━━━━ 3/3 4.5s/it 13.4s7.5s0s
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.2s/it 1.2s
                   all          6         13      0.949      0.692      0.74
7      0.643

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      16/50         0G     0.5609     0.9731     0.8897         27        64
0: 100% ━━━━━━━━━━━━ 3/3 5.0s/it 14.9s<10.0s
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 5.1s/it 5.1s
                   all          6         13       0.89      0.622      0.73
1      0.598

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      17/50         0G      0.503     0.9207     0.8647         34        64
0: 100% ━━━━━━━━━━━━ 3/3 4.8s/it 14.4s7.4s3s
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.3s/it 1.3s
                   all          6         13          1      0.534      0.71
1      0.597

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      18/50         0G     0.5257     0.8771     0.8793         30        64
0: 100% ━━━━━━━━━━━━ 3/3 4.3s/it 13.0s8.1ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.4s/it 1.4s
                   all          6         13          1      0.534      0.71
1      0.597

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      19/50         0G     0.5049     0.9473     0.9174         16        64
0: 100% ━━━━━━━━━━━━ 3/3 7.4s/it 22.2s<16.9s
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 5.9s/it 5.9s
                   all          6         13       0.85      0.538      0.71
4      0.647

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
```

```
e
      20/50         0G      0.558     0.8936     0.9059         33         64
0: 100% ━━━━━━━━━━━━ 3/3 5.9s/it 17.8s9.0s7s
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.2s/it 1.2s
                   all          6         13      0.875      0.538        0.7
2      0.638

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      21/50         0G     0.5113      0.909     0.8385         33         64
0: 100% ━━━━━━━━━━━━ 3/3 5.8s/it 17.4s9.2s7s
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.2s/it 1.2s
                   all          6         13      0.875      0.538        0.7
2      0.638

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      22/50         0G     0.5267     0.8574     0.8739         38         64
0: 100% ━━━━━━━━━━━━ 3/3 4.2s/it 12.6s6.9ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.3s/it 1.3s
                   all          6         13      0.886      0.597       0.78
8      0.667

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      23/50         0G     0.4453     0.7831     0.8871         33         64
0: 100% ━━━━━━━━━━━━ 3/3 4.0s/it 12.0s7.8ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.5s/it 1.5s
                   all          6         13      0.886      0.597       0.78
8      0.667

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      24/50         0G     0.4899     0.8187     0.9099         25         64
0: 100% ━━━━━━━━━━━━ 3/3 4.9s/it 14.7s9.3s9s
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.3s/it 1.3s
                   all          6         13      0.838      0.615       0.78
1      0.704

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      25/50         0G     0.4742     0.8558     0.9132         38         64
0: 100% ━━━━━━━━━━━━ 3/3 4.4s/it 13.1s8.3ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.7s/it 1.7s
                   all          6         13      0.838      0.615       0.78
1      0.704

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      26/50         0G      0.511     0.8602     0.8774         36         64
```

```
0: 100% ━━━━━━━━━━━━ 3/3 4.3s/it 13.0s7.7ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.1s/it 1.1s
                   all          6         13      0.986      0.615      0.77
6      0.694

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      27/50         0G     0.5453     0.8329     0.9092         39        64
0: 100% ━━━━━━━━━━━━ 3/3 4.5s/it 13.5s8.4ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.4s/it 1.4s
                   all          6         13      0.986      0.615      0.77
6      0.694

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      28/50         0G     0.5509     0.8039     0.8911         47        64
0: 100% ━━━━━━━━━━━━ 3/3 4.3s/it 13.0s8.4ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.1s/it 1.1s
                   all          6         13      0.776      0.802      0.84
7      0.754

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      29/50         0G     0.5394     0.9353     0.9615         51        64
0: 100% ━━━━━━━━━━━━ 3/3 4.0s/it 12.1s7.9ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.2s/it 1.2s
                   all          6         13      0.776      0.802      0.84
7      0.754

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      30/50         0G     0.4838     0.7988     0.9096         22        64
0: 100% ━━━━━━━━━━━━ 3/3 5.0s/it 15.0s9.5s4s
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.5s/it 1.5s
                   all          6         13      0.776      0.802      0.84
7      0.754

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      31/50         0G     0.4565     0.7598     0.8407         42        64
0: 100% ━━━━━━━━━━━━ 3/3 5.0s/it 14.9s8.3s4s
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.7s/it 1.7s
                   all          6         13      0.896      0.846        0.
9      0.796

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      32/50         0G     0.4366     0.8019     0.8837         33        64
0: 100% ━━━━━━━━━━━━ 3/3 4.4s/it 13.3s9.0ss
                 Class     Images  Instances      Box(P          R      mAP5
```

```
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.2s/it 1.2s
                   all          6         13      0.896      0.846        0.
9      0.796

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      33/50         0G     0.5196     0.8048     0.9078         25         64
0: 100% ━━━━━━━━━━━━ 3/3 4.3s/it 12.8s7.8ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.5s/it 1.5s
                   all          6         13      0.884      0.846      0.90
5      0.813

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      34/50         0G     0.4521     0.7759     0.8686         22         64
0: 100% ━━━━━━━━━━━━ 3/3 4.1s/it 12.4s7.7ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.2s/it 1.2s
                   all          6         13      0.884      0.846      0.90
5      0.813

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      35/50         0G     0.4424     0.7553     0.8866         42         64
0: 100% ━━━━━━━━━━━━ 3/3 4.7s/it 14.2s9.4ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.3s/it 1.3s
                   all          6         13      0.884      0.846      0.90
5      0.813

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      36/50         0G     0.4854     0.7801     0.8829         26         64
0: 100% ━━━━━━━━━━━━ 3/3 4.1s/it 12.3s8.2ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.4s/it 1.4s
                   all          6         13      0.908      0.923      0.91
9      0.843

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      37/50         0G     0.4914     0.7061      0.875         36         64
0: 100% ━━━━━━━━━━━━ 3/3 4.0s/it 12.0s7.2ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.4s/it 1.4s
                   all          6         13      0.908      0.923      0.91
9      0.843

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      38/50         0G     0.4331      0.755     0.8677         27         64
0: 100% ━━━━━━━━━━━━ 3/3 4.1s/it 12.4s8.3ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 6.6s/it 6.6s
                   all          6         13      0.908      0.923      0.91
```

```
9      0.843

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      39/50         0G     0.4639     0.7978     0.8631         35        64
0: 100% ━━━━━━━━━━━━ 3/3 6.7s/it 20.2s7.5s6s
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.3s/it 1.3s
                   all          6         13      0.919      0.923      0.91
8      0.847

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      40/50         0G     0.4252     0.7472      0.851         23        64
0: 100% ━━━━━━━━━━━━ 3/3 3.9s/it 11.7s7.3ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.2s/it 1.2s
                   all          6         13      0.919      0.923      0.91
8      0.847
Closing dataloader mosaic

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      41/50         0G     0.4577      1.149     0.8685         23        64
0: 100% ━━━━━━━━━━━━ 3/3 4.3s/it 12.9s7.6ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.1s/it 1.1s
                   all          6         13      0.841      0.923      0.91
2      0.852

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      42/50         0G     0.3805      1.048     0.8508         14        64
0: 100% ━━━━━━━━━━━━ 3/3 3.5s/it 10.6s6.6ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.1s/it 1.1s
                   all          6         13      0.841      0.923      0.91
2      0.852

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      43/50         0G     0.3647      1.034     0.8822          8        64
0: 100% ━━━━━━━━━━━━ 3/3 3.4s/it 10.3s6.3ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.1s/it 1.1s
                   all          6         13      0.841      0.923      0.91
2      0.852

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      44/50         0G     0.4105     0.8678     0.9027         18        64
0: 100% ━━━━━━━━━━━━ 3/3 3.7s/it 11.2s7.3ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.8s/it 1.8s
                   all          6         13        0.9      0.923      0.91
6       0.81
```

```
      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      45/50         0G     0.4051      0.937     0.8319         25        64
0: 100% ━━━━━━━━━━━━ 3/3 3.9s/it 11.6s7.3ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.1s/it 1.1s
                   all          6         13        0.9      0.923      0.91
6       0.81

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      46/50         0G     0.4131     0.9182     0.8721         15        64
0: 100% ━━━━━━━━━━━━ 3/3 3.7s/it 11.0s7.0ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.2s/it 1.2s
                   all          6         13        0.9      0.923      0.91
6       0.81

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      47/50         0G     0.3699      0.896     0.8535         11        64
0: 100% ━━━━━━━━━━━━ 3/3 4.4s/it 13.2s7.9ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.1s/it 1.1s
                   all          6         13      0.922      0.909      0.92
1      0.803

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      48/50         0G     0.3472     0.8415     0.8116         14        64
0: 100% ━━━━━━━━━━━━ 3/3 3.9s/it 11.6s7.5ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.1s/it 1.1s
                   all          6         13      0.922      0.909      0.92
1      0.803

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      49/50         0G      0.308     0.8612      0.786         29        64
0: 100% ━━━━━━━━━━━━ 3/3 4.4s/it 13.1s8.9ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.4s/it 1.4s
                   all          6         13      0.923      0.923      0.91
6      0.793

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
      50/50         0G     0.4045      1.012     0.8502         14        64
0: 100% ━━━━━━━━━━━━ 3/3 3.9s/it 11.8s6.8ss
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 1.1s/it 1.1s
                   all          6         13      0.923      0.923      0.91
6      0.793

50 epochs completed in 0.215 hours.
```

```
Optimizer stripped from /home/saidul/Desktop/4-2/Deep Learning/runs/detect/m
onitor_detector/weights/last.pt, 6.2MB
Optimizer stripped from /home/saidul/Desktop/4-2/Deep Learning/runs/detect/m
onitor_detector/weights/best.pt, 6.2MB

Validating /home/saidul/Desktop/4-2/Deep Learning/runs/detect/monitor_detect
or/weights/best.pt...
Ultralytics 8.4.51 🚀 Python-3.12.3 torch-2.12.0+cu130 CPU (Intel Core i7-86
65U 1.90GHz)
Model summary (fused): 73 layers, 3,005,843 parameters, 0 gradients, 8.1 GFL
OPs
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 2.6s/it 2.6s
                   all          6         13      0.922      0.915      0.91
7      0.851
Speed: 2.5ms preprocess, 341.4ms inference, 0.0ms loss, 60.9ms postprocess p
er image
Results saved to /home/saidul/Desktop/4-2/Deep Learning/runs/detect/monitor_
detector
Training completed!
```

In [4]:

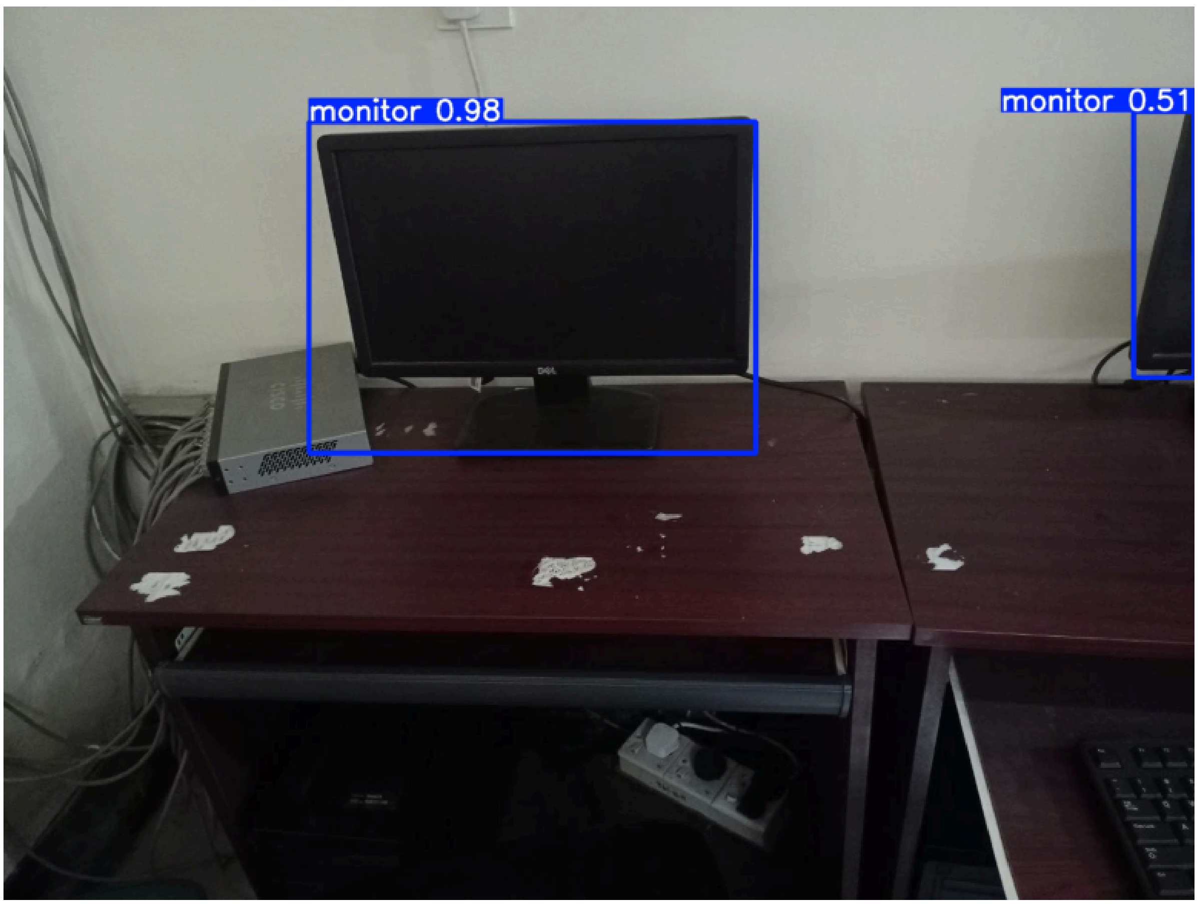
```python
# Evaluate on test set
metrics = model.val()

# Run inference on test images
results = model.predict(source="/home/saidul/Desktop/4-2/Deep Learning/data/

print("mAP50:", metrics.box.map50)
print("mAP50-95:", metrics.box.map)
```

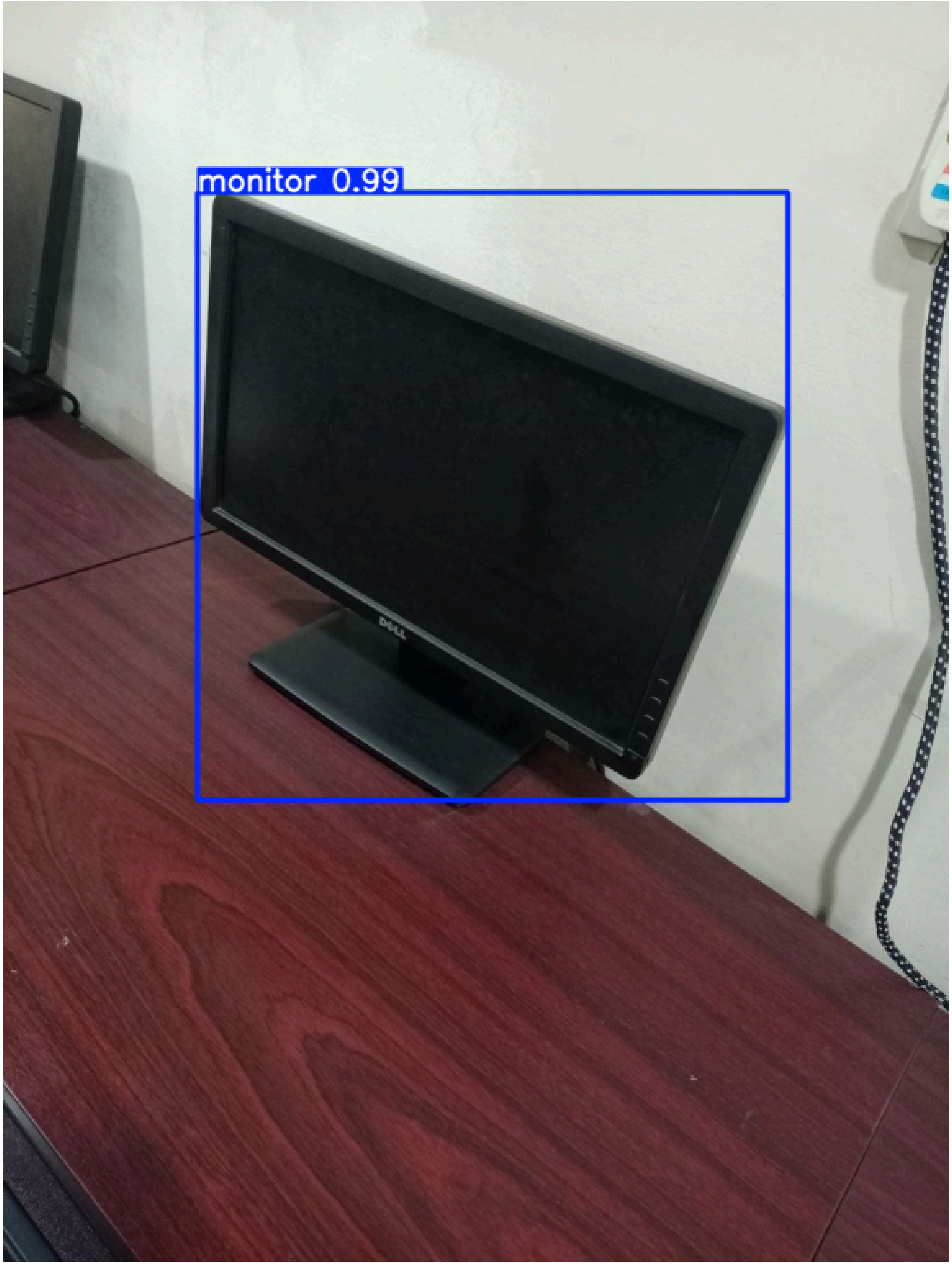
```
Ultralytics 8.4.51 🚀 Python-3.12.3 torch-2.12.0+cu130 CPU (Intel Core i7-86
65U 1.90GHz)
val: Fast image access ✅ (ping: 0.0±0.0 ms, read: 1588.6±435.2 MB/s, size:
113.0 KB)
val: Scanning /home/saidul/Desktop/4-2/Deep Learning/data/monitor_detection/
labels/val.cache... 6 images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 6/
6 1.4Mit/s 0.0s
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 1/1 2.6s/it 2.6s
                   all          6         13      0.841      0.923      0.91
2      0.852
Speed: 5.7ms preprocess, 352.4ms inference, 0.0ms loss, 22.7ms postprocess p
er image
Results saved to /home/saidul/Desktop/4-2/Deep Learning/runs/detect/val-2

image 1/5 /home/saidul/Desktop/4-2/Deep Learning/data/monitor_detection/imag
es/test/photo_22_2026-05-17_14-26-44_jpg.rf.XRpRV4KvI7zSOj2h0qTe.jpg: 480x64
0 2 monitors, 302.4ms
image 2/5 /home/saidul/Desktop/4-2/Deep Learning/data/monitor_detection/imag
es/test/photo_25_2026-05-17_14-26-44_jpg.rf.b5EpI0LpzIoQJIzBa78u.jpg: 640x48
0 2 monitors, 182.3ms
image 3/5 /home/saidul/Desktop/4-2/Deep Learning/data/monitor_detection/imag
es/test/photo_2_2026-05-17_14-26-44_jpg.rf.zJwVgbMk9QaF8l12diLR.jpg: 480x640
2 monitors, 219.3ms
image 4/5 /home/saidul/Desktop/4-2/Deep Learning/data/monitor_detection/imag
es/test/photo_31_2026-05-17_14-26-44_jpg.rf.lvF7STvxKJIs1xFqm1qD.jpg: 640x48
0 3 monitors, 216.6ms
image 5/5 /home/saidul/Desktop/4-2/Deep Learning/data/monitor_detection/imag
es/test/photo_4_2026-05-17_14-26-44_jpg.rf.2GElnR4rEvENvfnh5YfE.jpg: 640x480
3 monitors, 187.9ms
Speed: 8.1ms preprocess, 221.7ms inference, 1.5ms postprocess per image at s
hape (1, 3, 640, 480)
Results saved to /home/saidul/Desktop/4-2/Deep Learning/runs/detect/predict
mAP50: 0.9121689895470384
mAP50-95: 0.8519945799457996
```

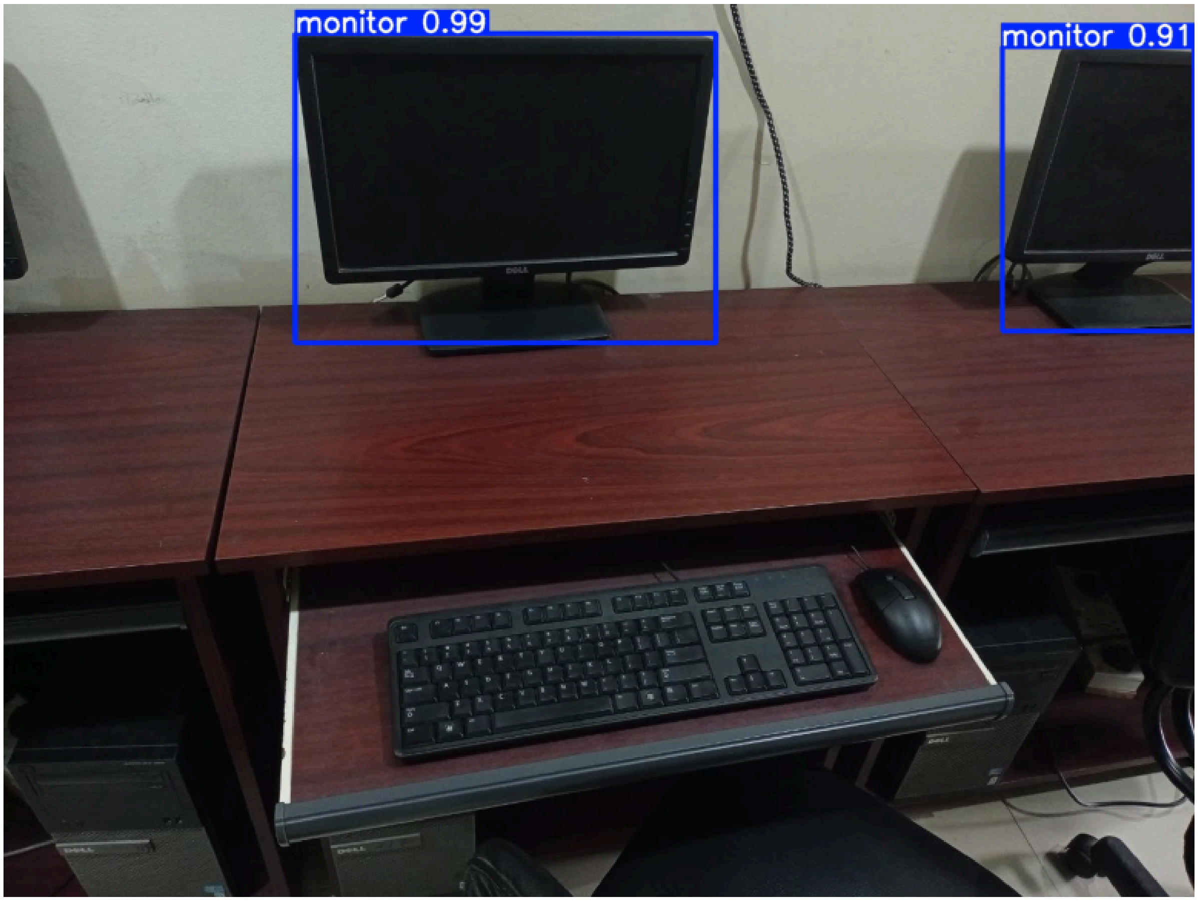
In [14]:

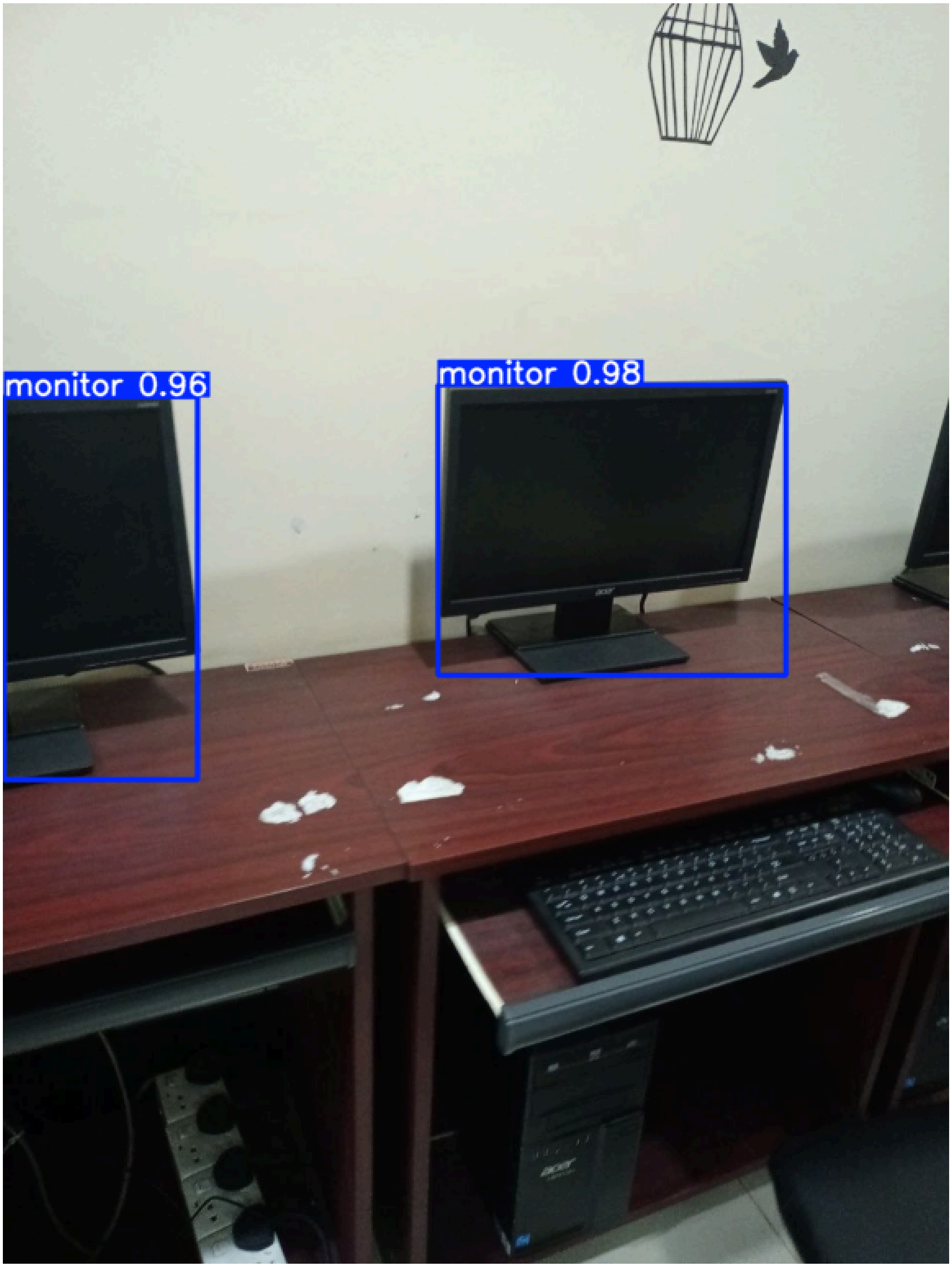
```python
import cv2
import matplotlib.pyplot as plt

results = model.predict("/home/saidul/Desktop/4-2/Deep Learning/data/monitor

for img_result in results:
    annotated_img = img_result.plot()
    rgb_img = cv2.cvtColor(annotated_img, cv2.COLOR_BGR2RGB)

    plt.figure(figsize=(10, 10))
    plt.imshow(rgb_img)
    plt.axis('off')
    plt.show()
```

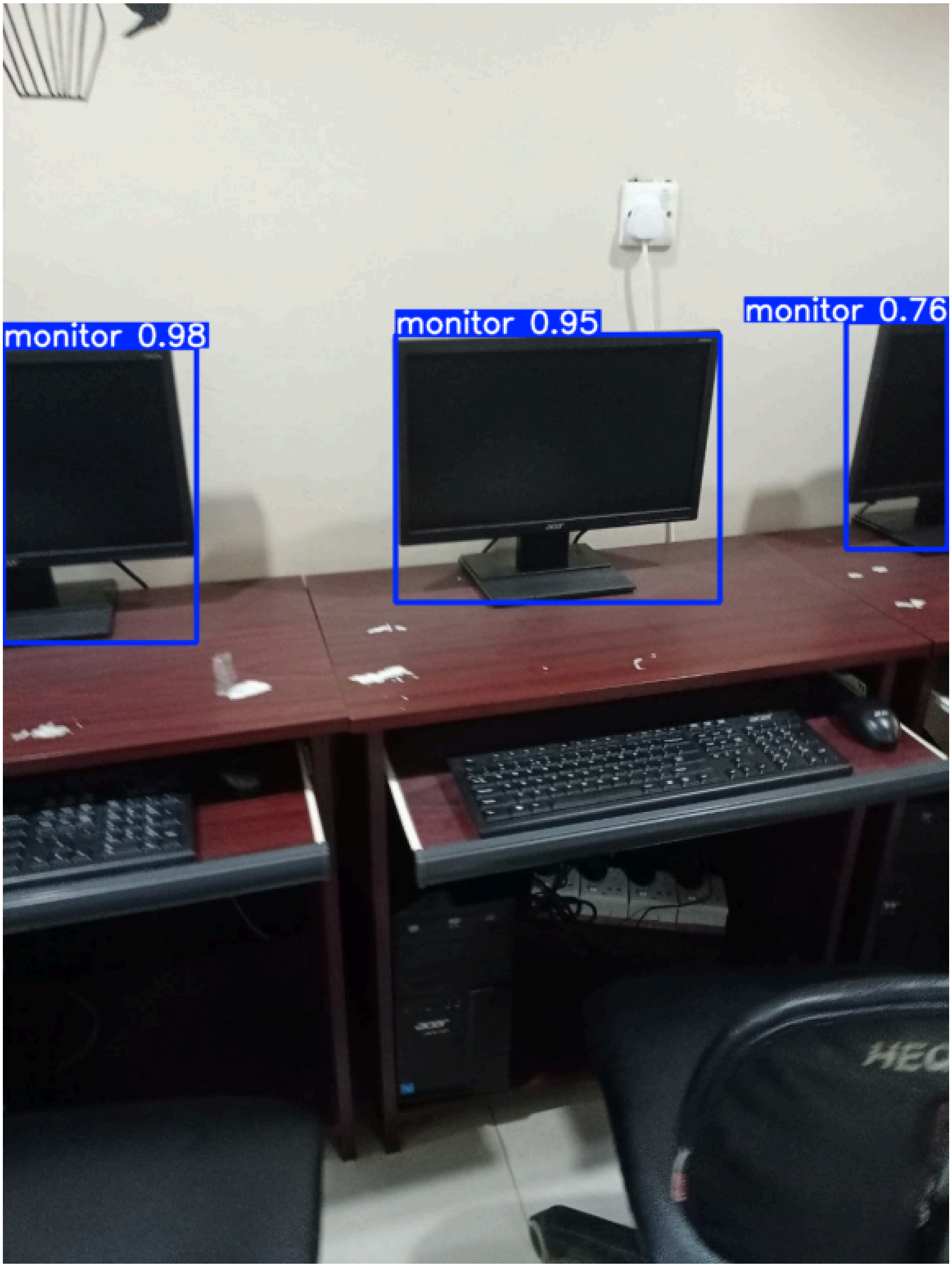
```
image 1/5 /home/saidul/Desktop/4-2/Deep Learning/data/monitor_detection/imag
es/test/photo_22_2026-05-17_14-26-44_jpg.rf.XRpRV4KvI7zSOj2h0qTe.jpg: 480x64
0 2 monitors, 132.2ms
image 2/5 /home/saidul/Desktop/4-2/Deep Learning/data/monitor_detection/imag
es/test/photo_25_2026-05-17_14-26-44_jpg.rf.b5EpI0LpzIoQJIzBa78u.jpg: 640x48
0 1 monitor, 95.5ms
image 3/5 /home/saidul/Desktop/4-2/Deep Learning/data/monitor_detection/imag
es/test/photo_2_2026-05-17_14-26-44_jpg.rf.zJwVgbMk9QaF8l12diLR.jpg: 480x640
2 monitors, 90.7ms
image 4/5 /home/saidul/Desktop/4-2/Deep Learning/data/monitor_detection/imag
es/test/photo_31_2026-05-17_14-26-44_jpg.rf.lvF7STvxKJIs1xFqm1qD.jpg: 640x48
0 2 monitors, 122.9ms
image 5/5 /home/saidul/Desktop/4-2/Deep Learning/data/monitor_detection/imag
es/test/photo_4_2026-05-17_14-26-44_jpg.rf.2GElnR4rEvENvfnh5YfE.jpg: 640x480
3 monitors, 174.6ms
Speed: 3.5ms preprocess, 123.2ms inference, 1.1ms postprocess per image at s
hape (1, 3, 640, 480)
Results saved to /home/saidul/Desktop/4-2/Deep Learning/runs/detect/predict
```




monitor 0.99


monitor 0.99
monitor 0.91


monitor 0.96
monitor 0.98


monitor 0.98
monitor 0.95
monitor 0.76

In [4]:

```
from ultralytics import YOLO
import cv2
import time

# Load video
video_path = "/home/saidul/Desktop/4-2/Deep Learning/data/Object_detection_v

# Models to compare
models = {
    "YOLOv8n": "yolov8n.pt",
    "YOLOv11n": "yolo11n.pt",
    "YOLOv12n": "yolo12n.pt"
}

results_summary = {}

for model_name, weight in models.items():
    print(f"\n Running {model_name}...")
    model = YOLO(weight)

    start_time = time.time()
    results = model.predict(
        source=video_path,
        save=True,           # Saves output video
        conf=0.25,
        iou=0.45,
        stream=True,
        verbose=False
    )
    end_time = time.time()

    total_time = end_time - start_time

    frame_count = 0
    for _ in results:
        frame_count += 1
    fps = frame_count / total_time if total_time > 0 else 0

    print(f"{model_name} finished in {total_time:.2f} seconds | Approx FPS: 
    results_summary[model_name] = {"time": total_time, "fps": fps}

print("\n=== Comparison Summary ===")
for name, data in results_summary.items():
    print(f"{name:10s} → Time: {data['time']:.2f}s | FPS: {data['fps']:.1f}"
```

```
 Running YOLOv8n...
Results saved to /home/saidul/Desktop/4-2/Deep Learning/runs/detect/predict-2
YOLOv8n finished in 0.03 seconds | Approx FPS: 14762.3

 Running YOLOv11n...
Results saved to /home/saidul/Desktop/4-2/Deep Learning/runs/detect/predict-3
YOLOv11n finished in 0.07 seconds | Approx FPS: 6462.7

 Running YOLOv12n...
Results saved to /home/saidul/Desktop/4-2/Deep Learning/runs/detect/predict-4
YOLOv12n finished in 0.05 seconds | Approx FPS: 9598.9

=== Comparison Summary ===
YOLOv8n    → Time: 0.03s | FPS: 14762.3
YOLOv11n   → Time: 0.07s | FPS: 6462.7
YOLOv12n   → Time: 0.05s | FPS: 9598.9
```